

AI Lab Assignment

Question 1: Water Jug Problem

Aim:

To solve the Water Jug problem using uninformed search.

Objectives:

- Understand state space search.
- Obtain exactly 2 gallons in the 4-gallon jug.

Solution:

Step	Action	State (4G,3G)
Initial	Both empty	(0,0)
1	Fill 3-gallon jug	(0,3)
2	Pour 3G into 4G	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4G until full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2G into 4G	(2,0)

Conclusion:

The goal state (2,0) is reached using state-space search.

Question 2: Mars Rover Intelligent Agent

Aim:

To study Mars Rover as an intelligent agent.

Objectives:

- Identify percepts.
- Analyze environment.
- List actions.
- Evaluate performance.
- Choose suitable architecture.

Percepts: Camera images, terrain, rocks, temperature, pressure, radiation, battery.

Operating Environment: Partially observable, stochastic, sequential, dynamic, continuous, single agent.

Actions: Move, turn, capture images, collect/analyze samples, avoid obstacles, transmit data, recharge.

Performance: Navigation, sample accuracy, energy efficiency, communication, mission success.

Suitable Architecture: Model-Based Goal-Based Learning Agent.

Reason: Learns, plans and adapts in uncertain environments.

Conclusion:

Mars Rover uses AI to autonomously explore Mars and accomplish scientific missions efficiently.

Question 3: 8-Queens Problem

Aim

To explore the search space using AI search strategies and formulate the problem components of the 8-Queens problem.

Objectives

1. To understand the state-space representation of the 8-Queens problem.
2. To formulate the problem using AI problem-solving components.
3. To apply search strategies for finding a valid solution.
4. To ensure that no two queens attack each other.

Problem Statement

Place **8 queens** on an **8×8 chessboard** such that **no two queens attack each other**. A queen can attack another queen horizontally, vertically, and diagonally.

Problem Formulation

State

A board configuration containing 0 to 8 queens.

Initial State

An empty chessboard with no queens placed.

Actions

Place one queen in a safe position in the next column.

Transition Model

Adding one queen to a safe square generates a new state.

Goal Test

Eight queens are placed such that:

- No two queens share the same row.
- No two queens share the same column.
- No two queens share the same diagonal.

Path Cost

Each queen placement has a cost of 1. The objective is to reach the goal with eight successful placements.

Search Strategy

- State Space Search
- Depth-First Search (DFS)
- Backtracking Search (most commonly used)

Algorithm (Backtracking)

1. Start with an empty chessboard.
2. Place a queen in the first safe position of column 1.
3. Move to the next column.
4. If no safe position exists, backtrack to the previous column.
5. Continue until all 8 queens are placed safely.

Result

A valid arrangement of 8 queens is obtained where none attacks another.

Conclusion

The 8-Queens problem demonstrates the use of **state-space search** and **backtracking**, enabling AI systems to efficiently solve constraint satisfaction problems.

Question 4: OLA Cab Booking Problem

Aim

To identify the suitable AI problem-solving agent for the OLA Cab booking system and write its pseudocode.

Objectives

1. To understand the working of a problem-solving agent.
2. To identify the suitable AI agent for cab booking.
3. To write the pseudocode for selecting a cab.
4. To understand AI-based decision making.

Problem Description

A customer wants to travel from one location to another using the **OLA Cab** application. The customer enters the pickup and destination locations and selects a preferred cab type (Mini, Micro, Sedan, Prime, Shared, etc.).

Type of AI Agent

Goal-Based Problem-Solving Agent

Reason

The agent:

- Accepts the user's destination as the goal.
- Searches available cab options.
- Compares fare, availability, and travel time.
- Selects the most suitable cab based on the user's preference.

Pseudocode

START

Input Pickup Location

Input Destination

Display Available Cab Types

Customer Selects Cab Type

IF Selected Cab is Available THEN

 Calculate Fare

 Confirm Booking

 Assign Driver

 Display Driver Details

ELSE

 Display "Cab Not Available"

 Suggest Alternative Cab Types

ENDIF

END

Working

1. User enters pickup and destination.
2. System checks available cabs.
3. User selects a preferred cab.
4. The system calculates the fare.

5. Booking is confirmed if the cab is available.
6. Otherwise, an alternative cab type is suggested.

Result

The Goal-Based Problem-Solving Agent successfully selects the appropriate cab and completes the booking process.

Experiment 5: Uniform Cost Search (UCS) – Least Cost Path

Aim

To determine the **least-cost path** from the starting warehouse **S** to the destination warehouse **G** using the **Uniform Cost Search (UCS)** algorithm.

Objectives

1. To understand the Uniform Cost Search (UCS) algorithm.
2. To find the minimum-cost path in a weighted graph.
3. To calculate the cumulative path cost at each step.
4. To identify the optimal route from source to destination.

Problem Understanding

The logistics company needs to transport packages from warehouse **S** to warehouse **G** with the **minimum travel cost**.

Given Graph

Edges and Costs:

Edge	Cost
S → A	1
S → G	12
A → B	3
A → C	1
B → D	3
C → D	1
C → G	2
D → G	3

Algorithm Used

Uniform Cost Search (UCS)

Principle:

Always expand the node with the **lowest cumulative path cost** first.

Solution

Step 1

Start at S

Open List:

Node	Cost
S	0

Expand S

Generated Nodes:

- A (1)
- G (12)

Step 2

Choose lowest cost node

Expand A (1)

Generated:

- $B = 1 + 3 = 4$
- $C = 1 + 1 = 2$

Open List

Node	Cost
C	2
B	4
G	12

Step 3

Expand C (2)

Generated:

- $D = 2 + 1 = 3$
- $G = 2 + 2 = 4$

Open List

Node	Cost
D	3
B	4
G	4
G	12

Step 4

Expand **D (3)**

Generated

- $G = 3 + 3 = 6$

Open List

Node	Cost
B	4
G	4
G	6
G	12

Step 5

Expand **G (4)**

Goal reached.

Least-Cost Path

$$\boxed{S \rightarrow A \rightarrow C \rightarrow G}$$

Total Cost Calculation

$$S \rightarrow A = 1$$

$$A \rightarrow C = 1$$

$$C \rightarrow G = 2$$

$$\text{Total Cost} = 1 + 1 + 2 = \boxed{4}$$

Result

- **Optimal Path:** $S \rightarrow A \rightarrow C \rightarrow G$
- **Minimum Cost:** 4

Conclusion

Uniform Cost Search (UCS) successfully finds the **least-cost path** by always expanding the node with the lowest cumulative cost. For the given delivery network, the optimal route is $S \rightarrow A \rightarrow C \rightarrow G$ with a **minimum total cost of 4**, making it the most efficient path for package transportation.